Machine Learning

Course Introduction

Dong-Kyu Chae

PI of the Data Intelligence Lab @HYU
Department of Computer Science
Hanyang University

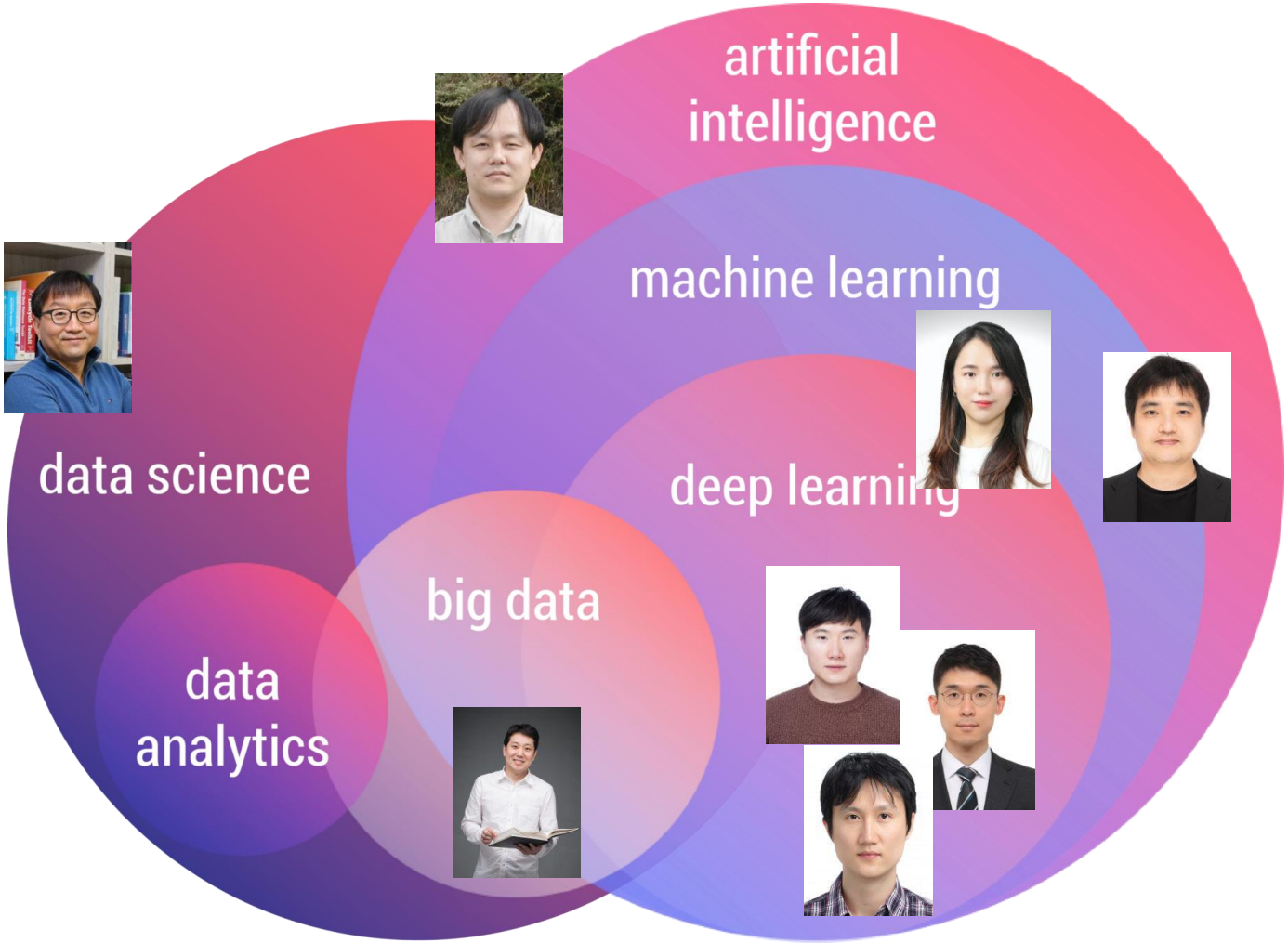
Data
Intelligence
LAB

Information

- **Expected audience**
  - Students who are trying to study machine learning for the **first** time
- **Required background**
  - Basic linear algebra
    - Vector, inner product, matrix, inverse matrix, transpose, …
  - Basic calculus
    - Partial derivative, continuous function, differentiable function, …
  - Basic probability & statistics
    - Probability distribution, conditional probability, sampling, Bayesian, …
- **Grades**
  - 40% of the total students will get A+ or A0

Overview

- **Terms**



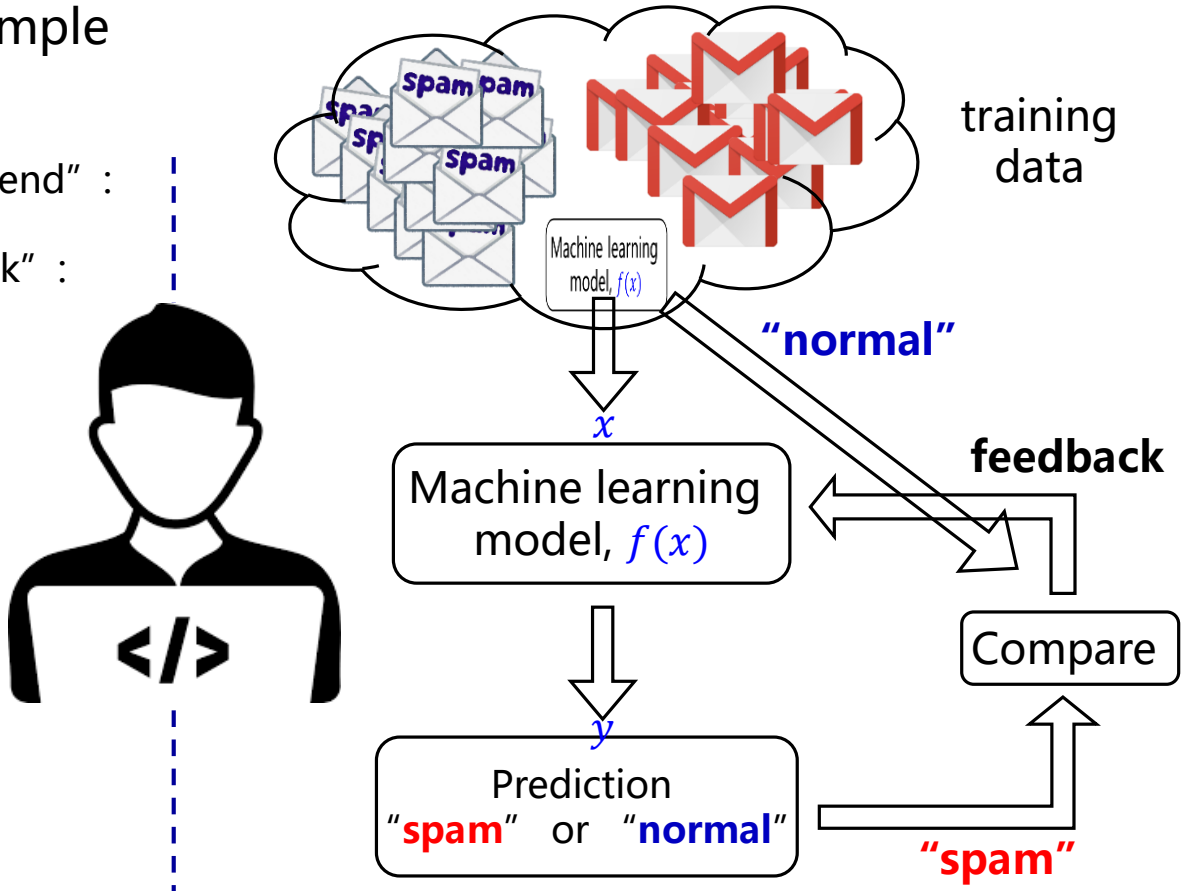
History of DS & AI

- **Association rules: Beer and diapers**

History of DS & AI

- **Machine learning: SVM, Random Forest, etc…**
  - Spam mail detector example

```
if email contains "bitcoin" and "send" :
  then mark spam;
if email contains "FREE" and "click" :
  then mark spam;
if email contains …
  then …
if email contains …
  then …
if email contains …
  then …
…..
```

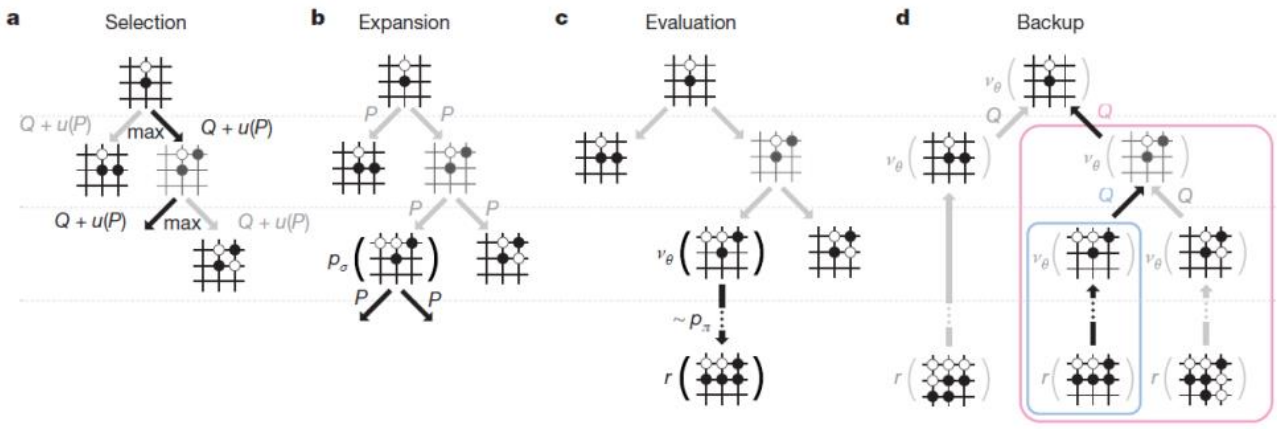


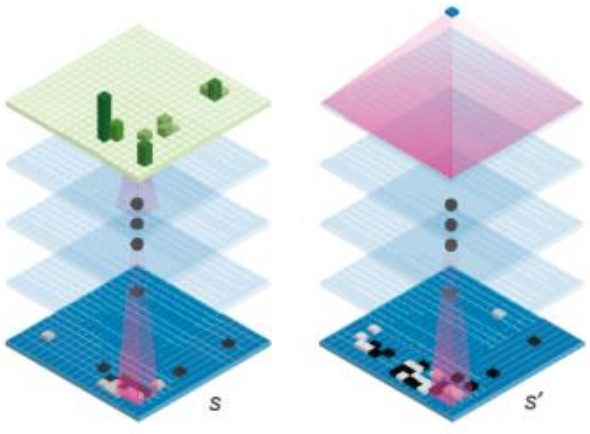
Traditional programming

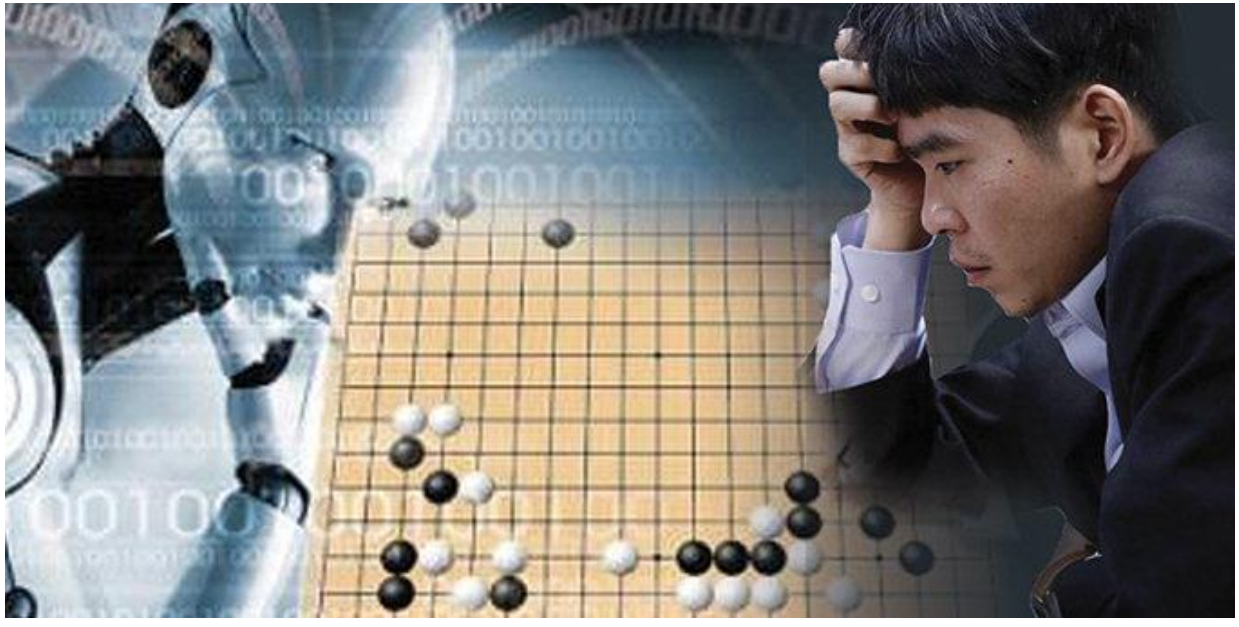
Machine learning

History of DS & AI

- **Deep learning (deep neural networks): human-level AI**







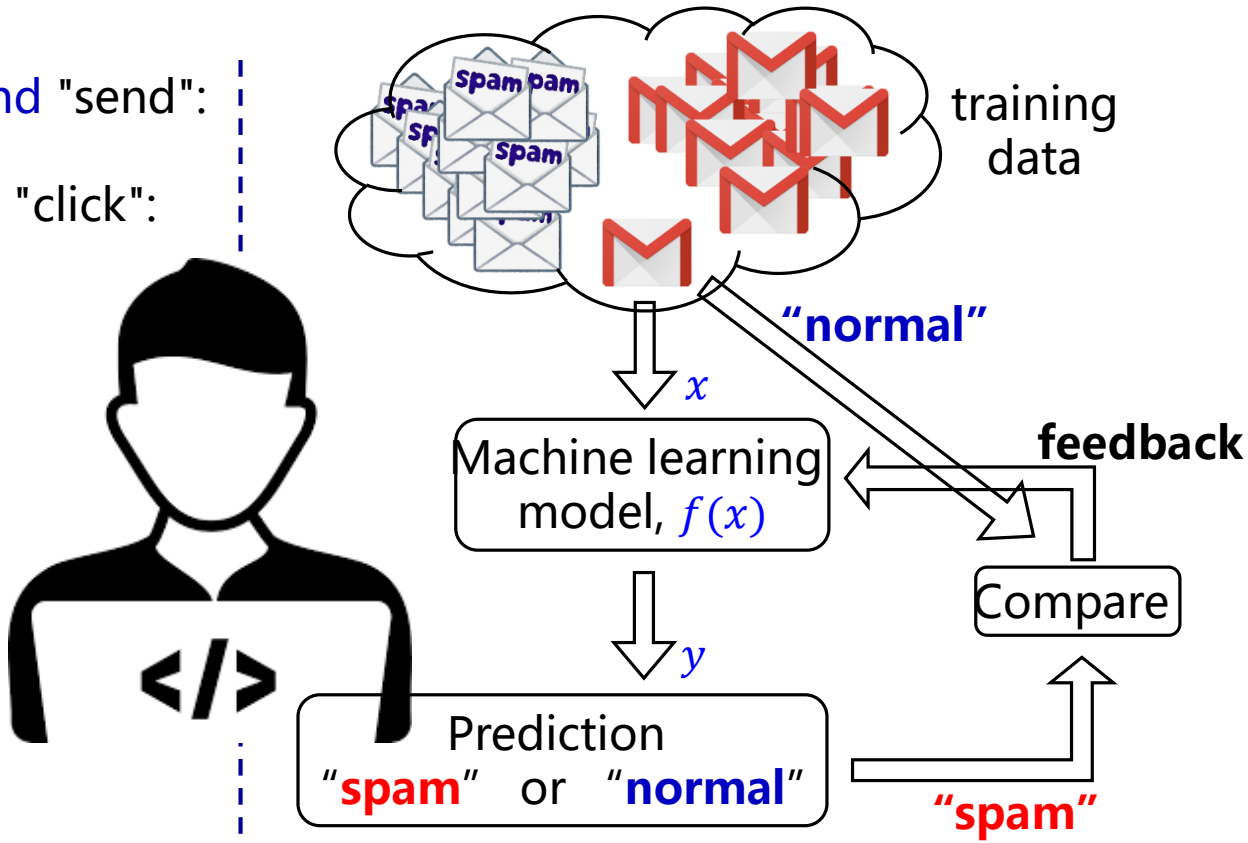
Machine Learning

- **What is machine learning?**
  - https://en.wikipedia.org/wiki/Machine_learning
  - Machine learning (ML) is the study of computer **algorithms** that **improve automatically** through **experience**.
  - Machine learning algorithms **build a mathematical model** based on sample data, known as "**training data**", in order to **make predictions** or decisions **without being explicitly programmed to do so**.

Machine Learning

- **Example: Making a spam mail detector**
  - Traditional programming: you need to **hard-code** the rules.
  - Machine learning: you can get **data-driven rules** without coding explicitly

```
if email contains "bitcoin" and "send":
  then mark spam;
if email contains "FREE" and "click":
  then mark spam;
if email contains ...
  then ...
if email contains ...
  then ...
if email contains ...
  then ...
.....
```



Traditional programming

Machine learning

Machine Learning

Process

1. Collect (or download) data
2. Pre-process data, define input features and output label
   - The values of features should be **interpretable** by computers

	# of words	# of attached files	# of links	# of malicious words	.....	spam
mail #1	256	0	3	7	.....	1 (Yes)
mail #2	56	1	0	3	.....	0 (No)
mail #3	24	1	0	1	.....	0
mail #4	672	0	0	0	.....	0
mail #5	67	2	4	3	.....	1
mail #6	48	0	2	6	.....	0
mail #7	79	1	3	8	.....	1
.....						

3. Select machine learning model
   - SVM, decision tree, regression, neural networks, …
4. **Train model with training data**

Machine Learning

- **What is a model?**
  - It is a **function** that maps inputs ($\mathbf{x}$) to an output ($\mathbf{y}$), **parameterized** by the set of **model parameters** Θ

    $\mathbf{x}$: email $\qquad$ $\mathbf{y}$ = spam
    - $\mathbf{x}_1$=<256, 0, 3, 7, …> $\quad$ $\mathbf{y}_1 = 1$
    - $\mathbf{x}_2$=< 56, 1, 0, 3, …> $\quad$ $\mathbf{y}_2 = 0$

    …..

  - $\mathbf{y} = f(\mathbf{x}; \Theta)$

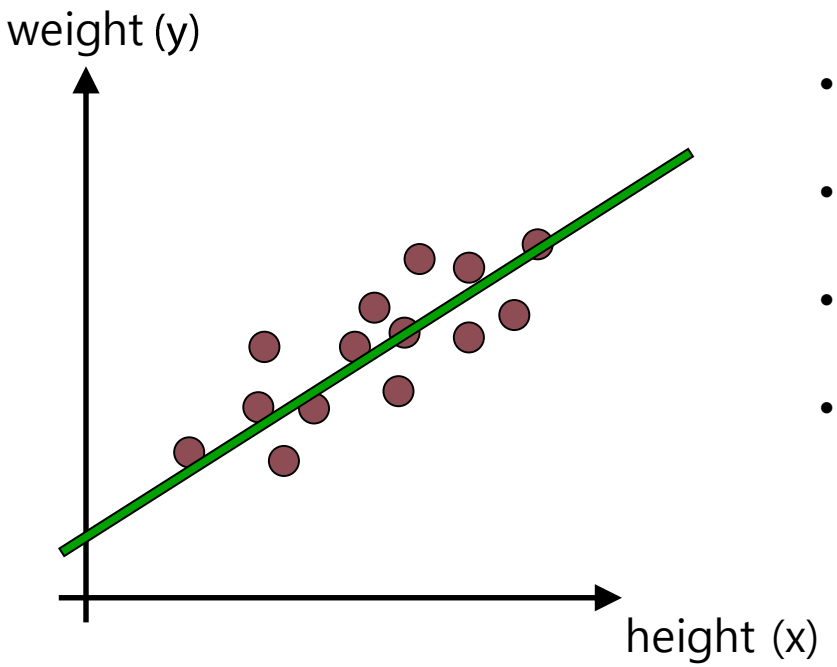
    ex) y=ax+b => Θ={a, b}

- **What is the meaning of "training a model" ?**
  - Finding a set of **optimal model parameters** that **best** map **x** to **y**

Machine Learning

A toy example

- Given a person's **height**, we want to build a model predicting his/her **weight**
- **Linear regression** was chosen as our model

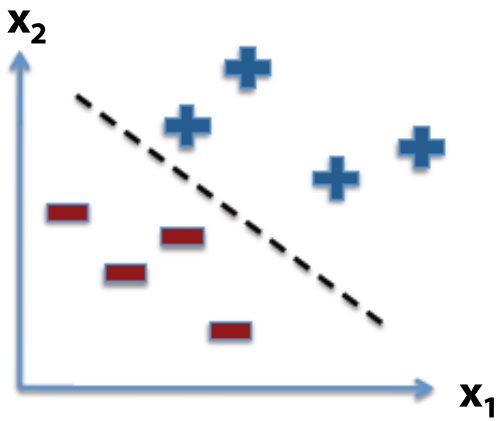


- Data: $(x_i, y_i) = (x_1, y_1), (x_2, y_2), \ldots (x_{14}, y_{14})$
- Model $f(x; \Theta) = ax + b$
- Model parameters $\Theta : \{a, b\}$
- Goal: finding the **optimal values** of **a** and **b**

Machine Learning

Classification models:

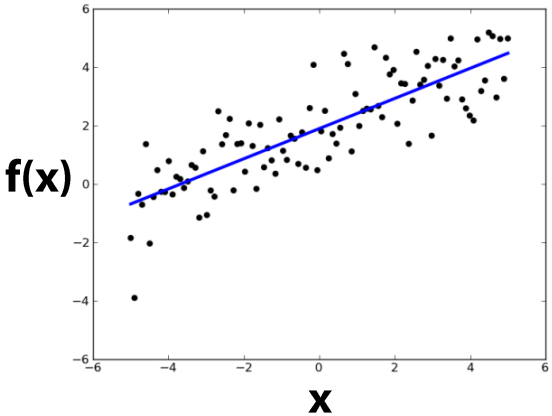
- Decision tree
- Bayesian classifier
- Logistic regression
- Neural network
- SVM
- …



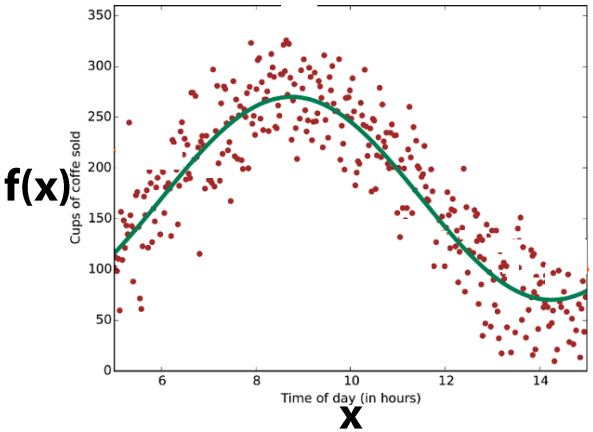
$\mathbf{f(x)} = ax_1+bx_2+c$

Regression models:

- Linear regression
- Neural network
- ….



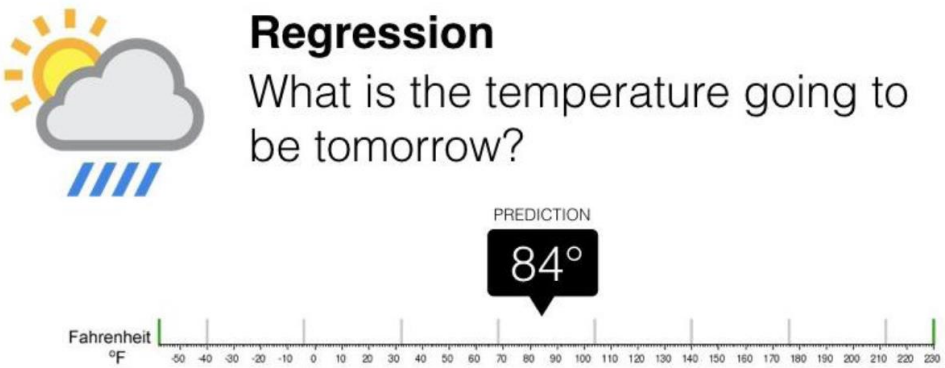
$\mathbf{f(x)} = ax+b$

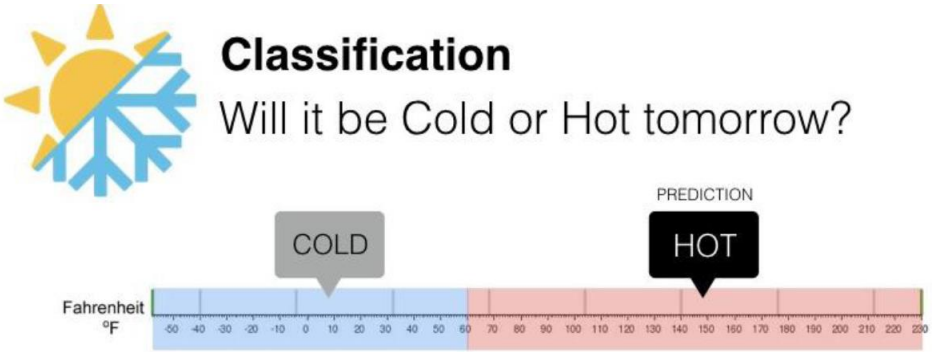


$\mathbf{f(x)} = ax^2+bx+c$

Category of Machine Learning

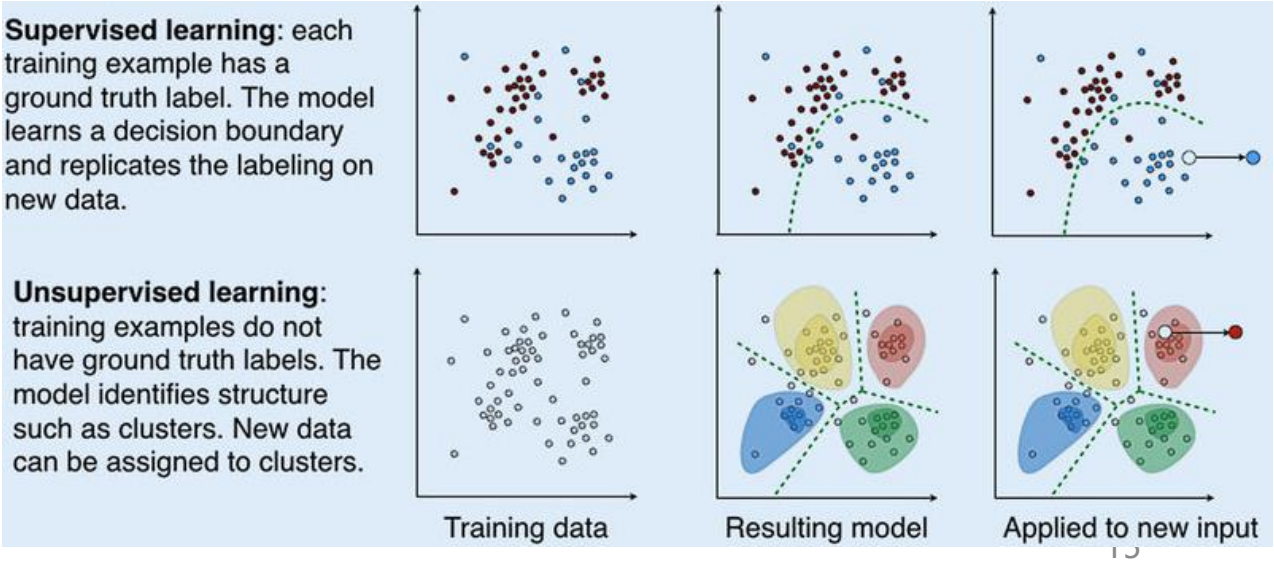
- **Supervised learning**
  - https://en.wikipedia.org/wiki/Supervised_learning
  - Supervised learning is the machine learning task of learning a function that maps an **input** to an **output** based on **example input-output pairs**. It infers a function from **labeled training data**.
- **Applications**
  - **Classification**: output is a class label
  - **Regression**: output is a value





Category of Machine Learning

- **Unsupervised learning**
  - https://en.wikipedia.org/wiki/Unsupervised_learning
  - Unsupervised learning is a type of machine learning that looks for **previously undetected patterns in a data** set with **no pre-existing labels** and **with a minimum of human supervision**.
- **Applications**
  - Density estimation
  - Clustering



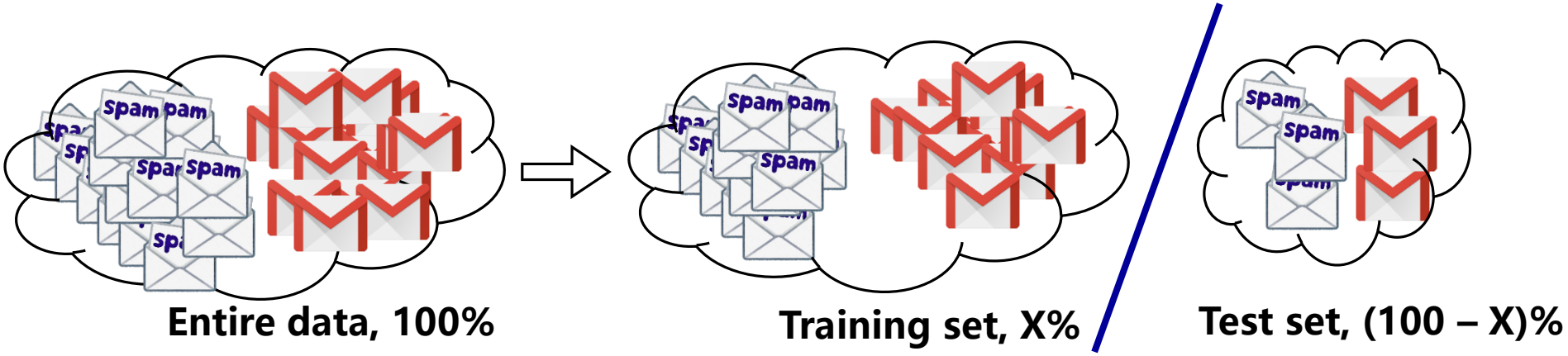
Accuracy of Machine Learning

- **Performance validation of classification**
  - Classification accuracy
    - After training a model with our training data
    - Collect 150 additional examples (Not from training data!)
    - Let the model predict their classes
    - Measure the accuracy by $= \frac{TP+TN}{150} * 100\%$
    - Also called as "**generalization performance**"

		Actual	
		Positive	Negative
Predicted	Positive	**True Positive (TP)**	**False Positive**
	Negative	**False Negative**	**True Negative (TN)**

Accuracy of Machine Learning

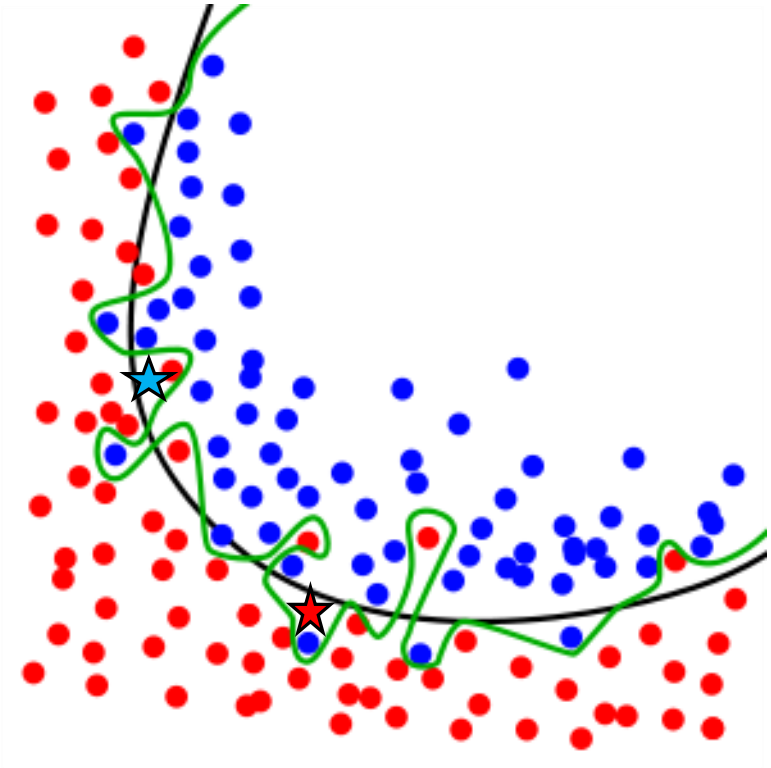
- **Performance validation of classification**
  - Splitting data into training and test sets.

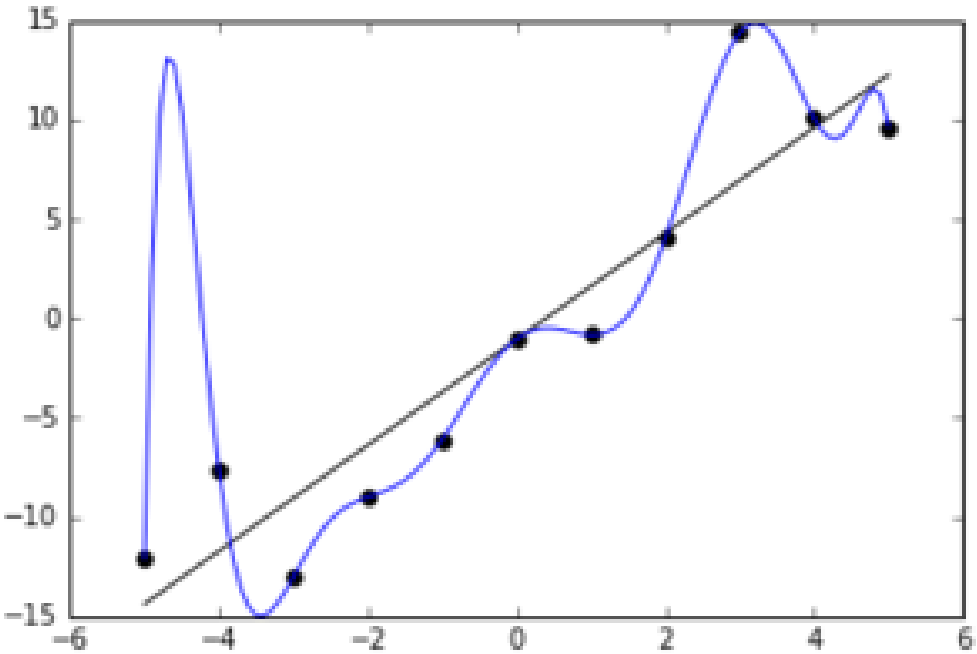


  - Model's training accuracy
    - Accuracy on classifying training data
    - It is just used for finding optimal model parameters
  - **Model's test accuracy**
    - Accuracy on classifying data that it has never seen before
    - It shows the model's generalization ability

Overfitting

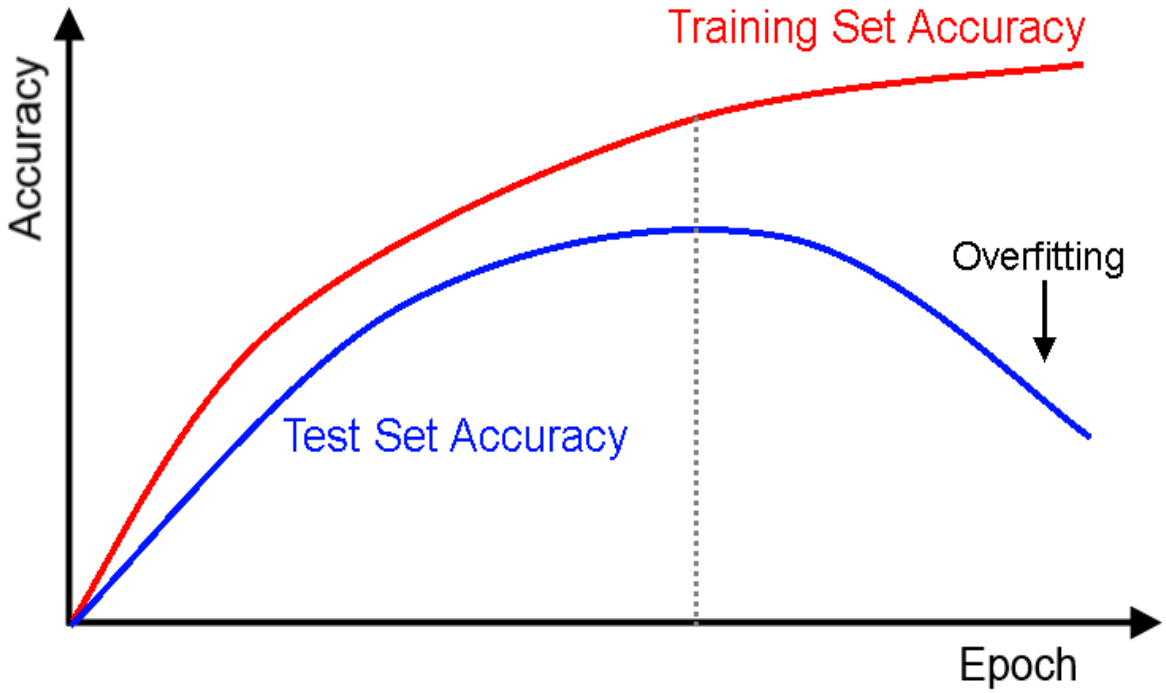
- **Result of a training that makes a model fit training data too closely.**
- **Perfect on training data, but poor on test data**
  - Bad generalization ability





Overfitting

- **Perfect on training data, but poor on test data**
  - Bad generalization ability



Topics to be Addressed

- **Optimization**
  - Linear optimization, analytic solution
  - Non-linear optimization, gradient descent
- **Linear classification**
  - Support vector machine
  - Logistic regression
- **Non-linear classification**
  - Artificial neural network
  - Back-propagation
- **Regularization**

Topics to be Addressed

- **Probabilistic Machine Learning**
  - Naïve Bayes classifier
  - Bayesian network
  - Expectation maximization
  - Variational inference
- **Unsupervised learning**
  - Clustering
  - Gaussian mixture models

Summary

- **Paradigm of machine learning**
  - Learning the data-driven decision boundary, instead of human programming
- **What is "training" or "learning" a model?**
  - It is a process to retrieve the **optimal model parameter** values that minimize the loss function => **an optimization problem!**
- **Types of machine learning**
  - Supervised, unsupervised, self-supervised, ...
- **Other topics**
  - Training / test set
  - Accuracy / overfitting
  - Evaluation of machine learning models

Thank You